Project Overview and Timeline

Fall 2025 Semester plan

JC Vaught

Big Picture

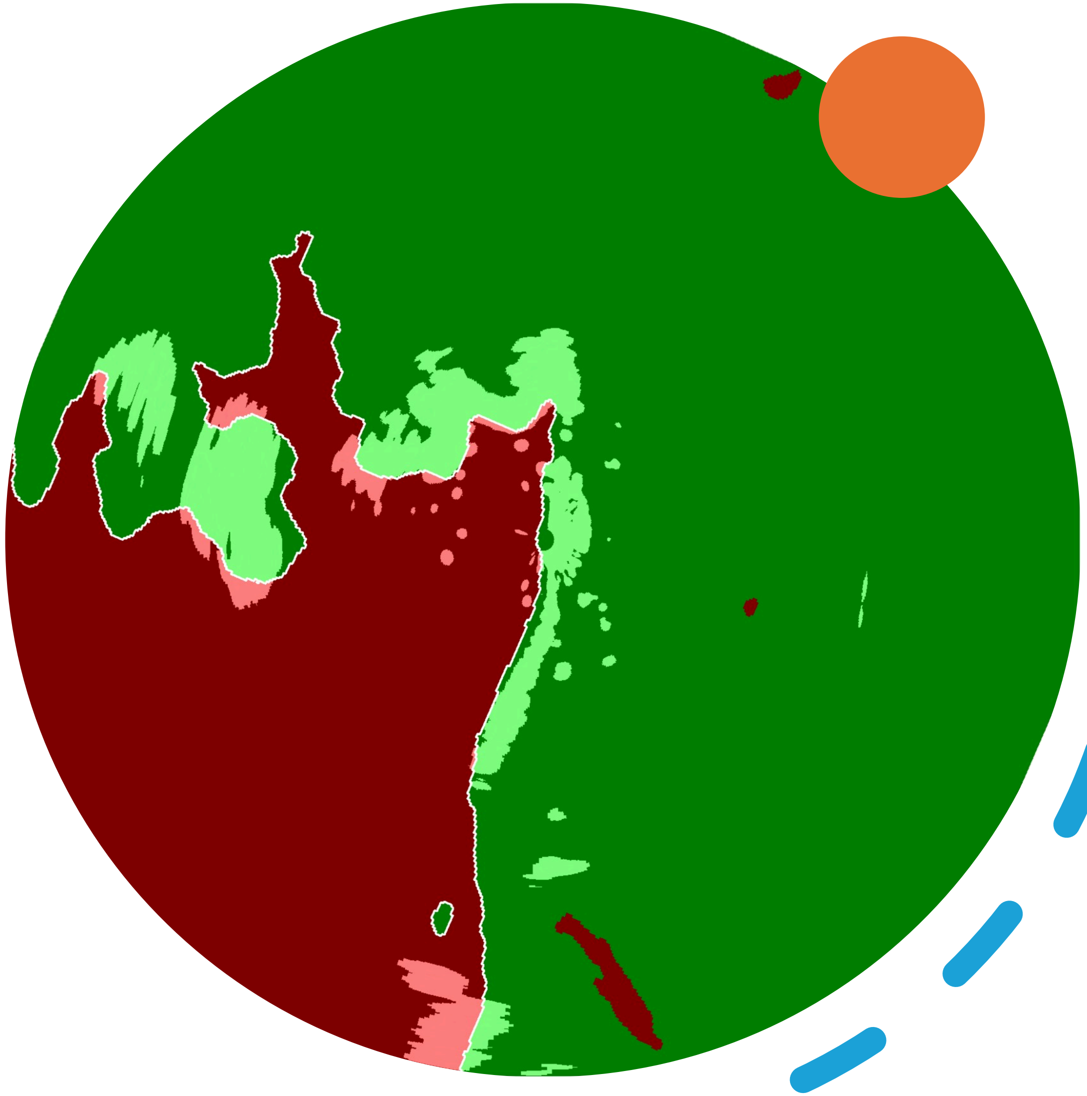
Many AI models exist to detect objects, but none of them are trained for small maritime objects.

Training AI to identify marine objects requires labeling thousands of images, a tedious and costly process

What does radar do for us?

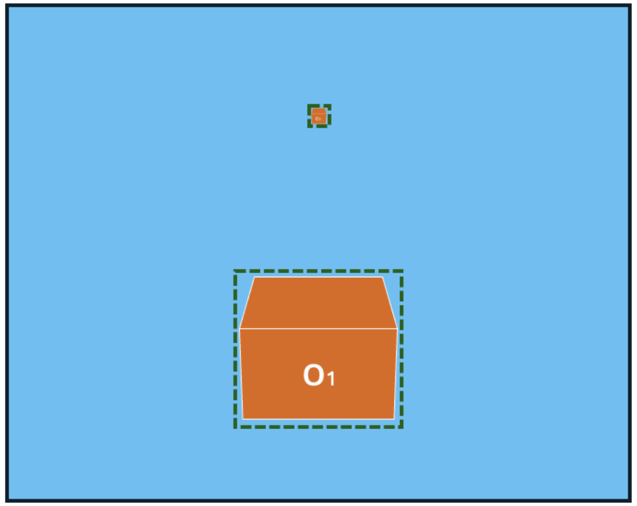
Main Goal: To create a system that automatically finds objects in camera images, making the data labeling process faster and more efficient.

The Core Idea: We will use one sensor (Radar) to "cue" another sensor (a Camera), telling it (or a person) where to look for objects.

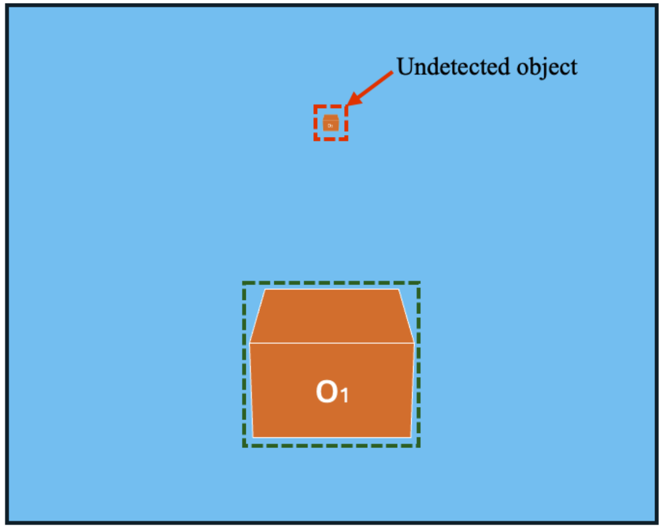


Our Tools

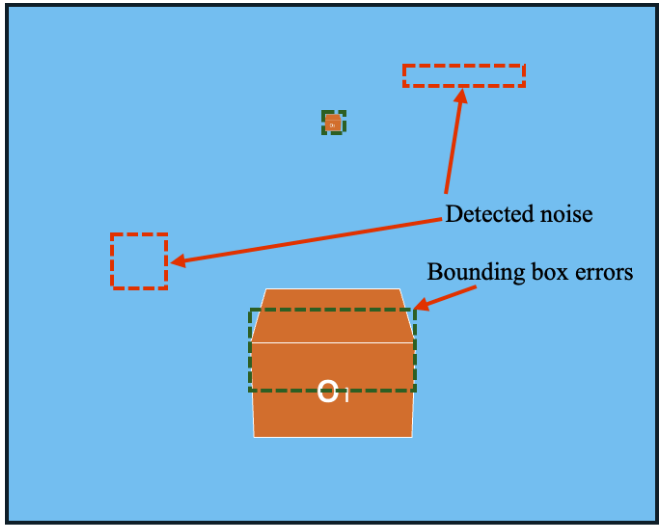
- Camera
  - Gives us a high-resolution, color picture of the world. Great for identifying what an object is (a boat, a buoy, etc.).
  - Weakness: Doesn't work well in fog, rain, or at night.
- Radar
  - Sends out radio waves and listens for echoes. Excellent at detecting objects at long distances, day or night, in any weather.
  - Weakness: Gives a low-resolution, "blob-like" picture. Can't easily tell a boat from a pylon.



(c) Sensor Fusion

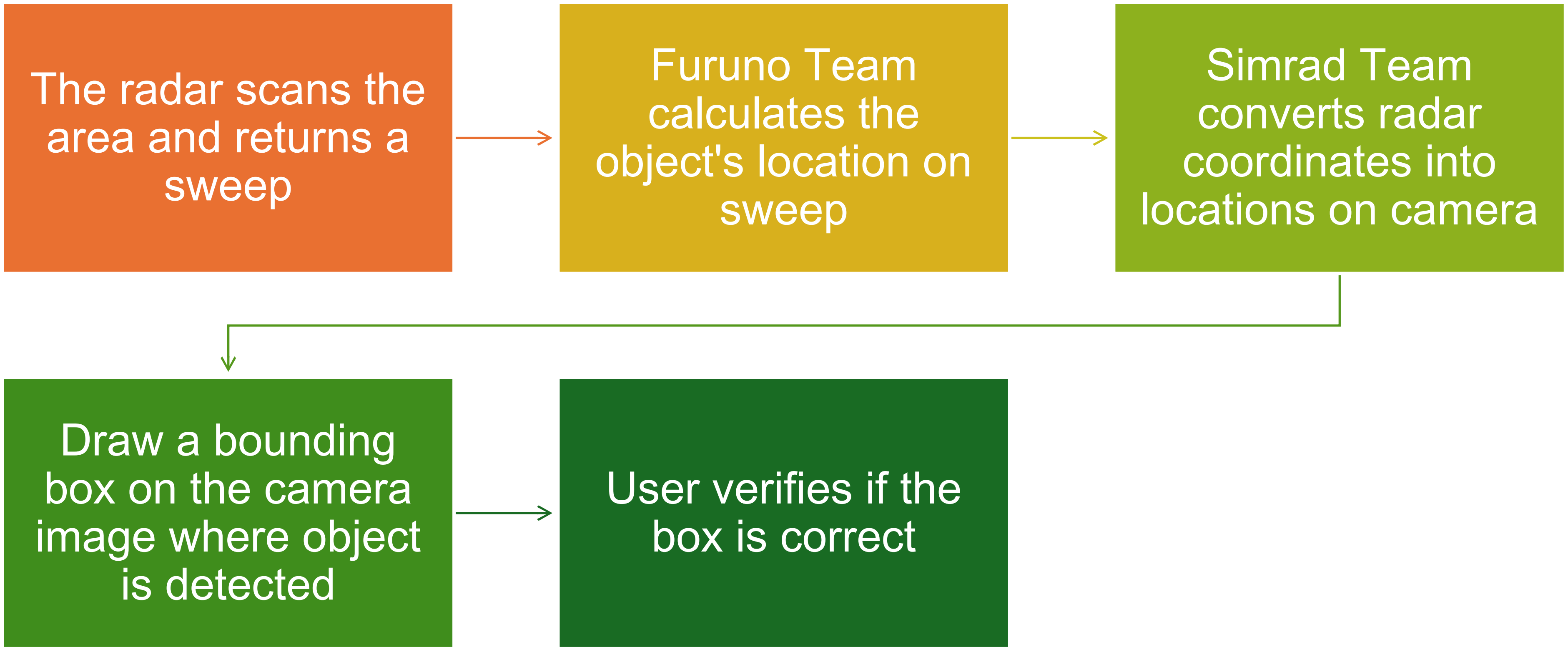


(a) Vision only



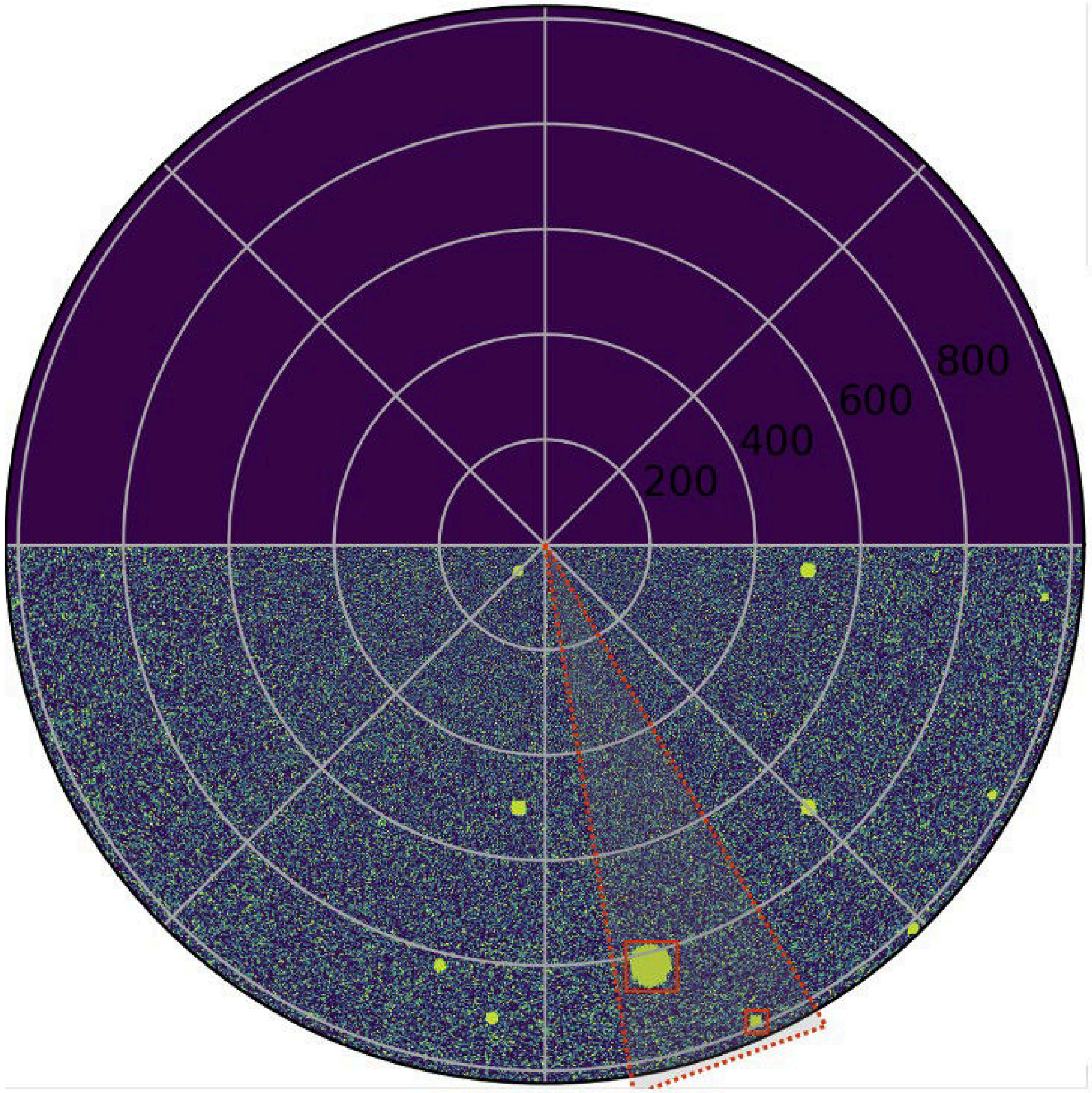
(b) Radar only

Auto-labelling



Method 2

(a) Radar Object Detection

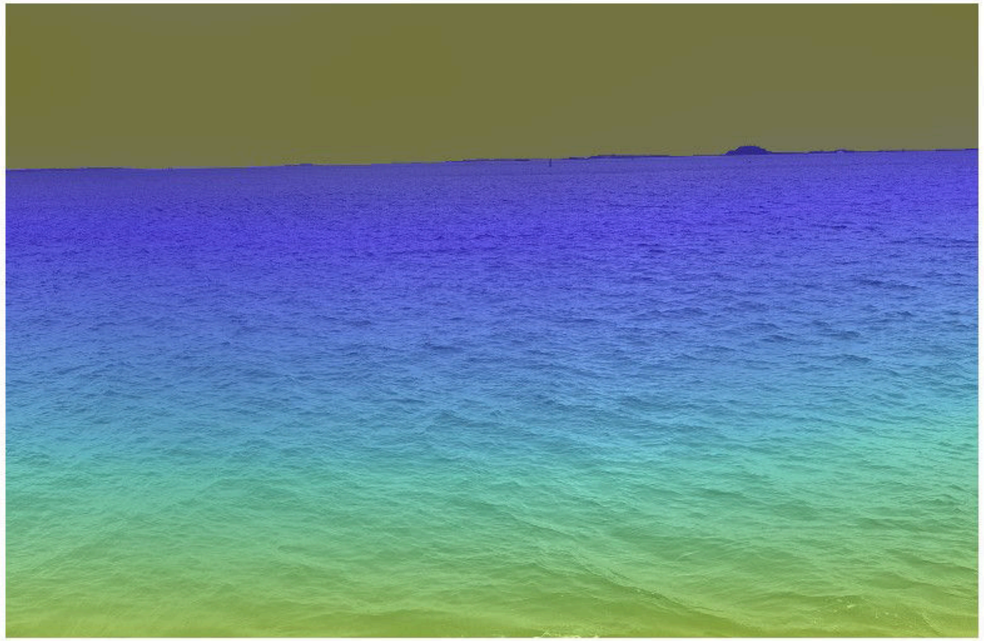


(b) Radar Overlay for Annotator

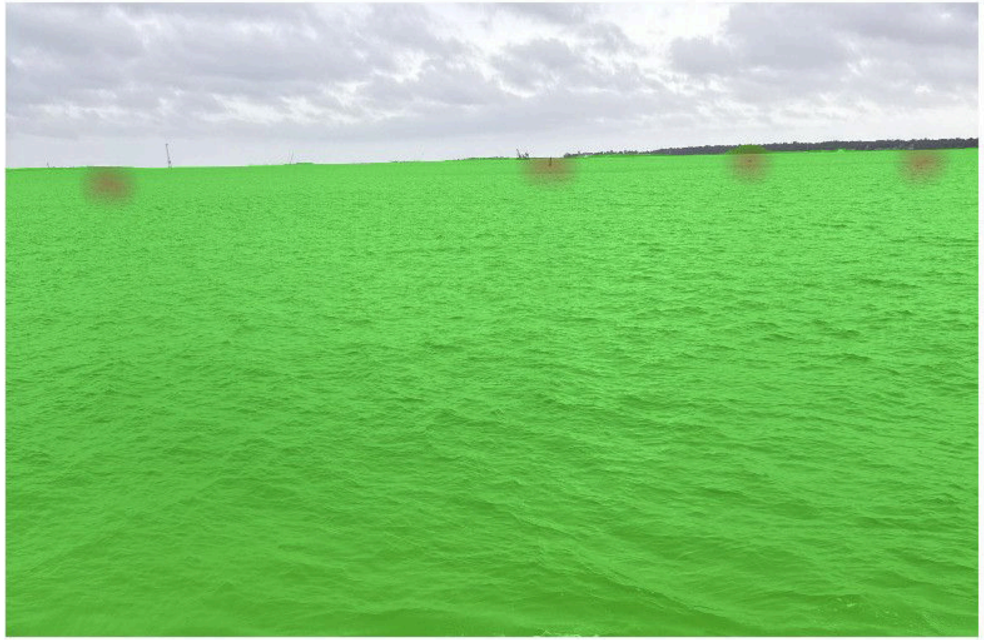
Method 1

(a) Original image

(b) Segmentation



(c) Depth Map

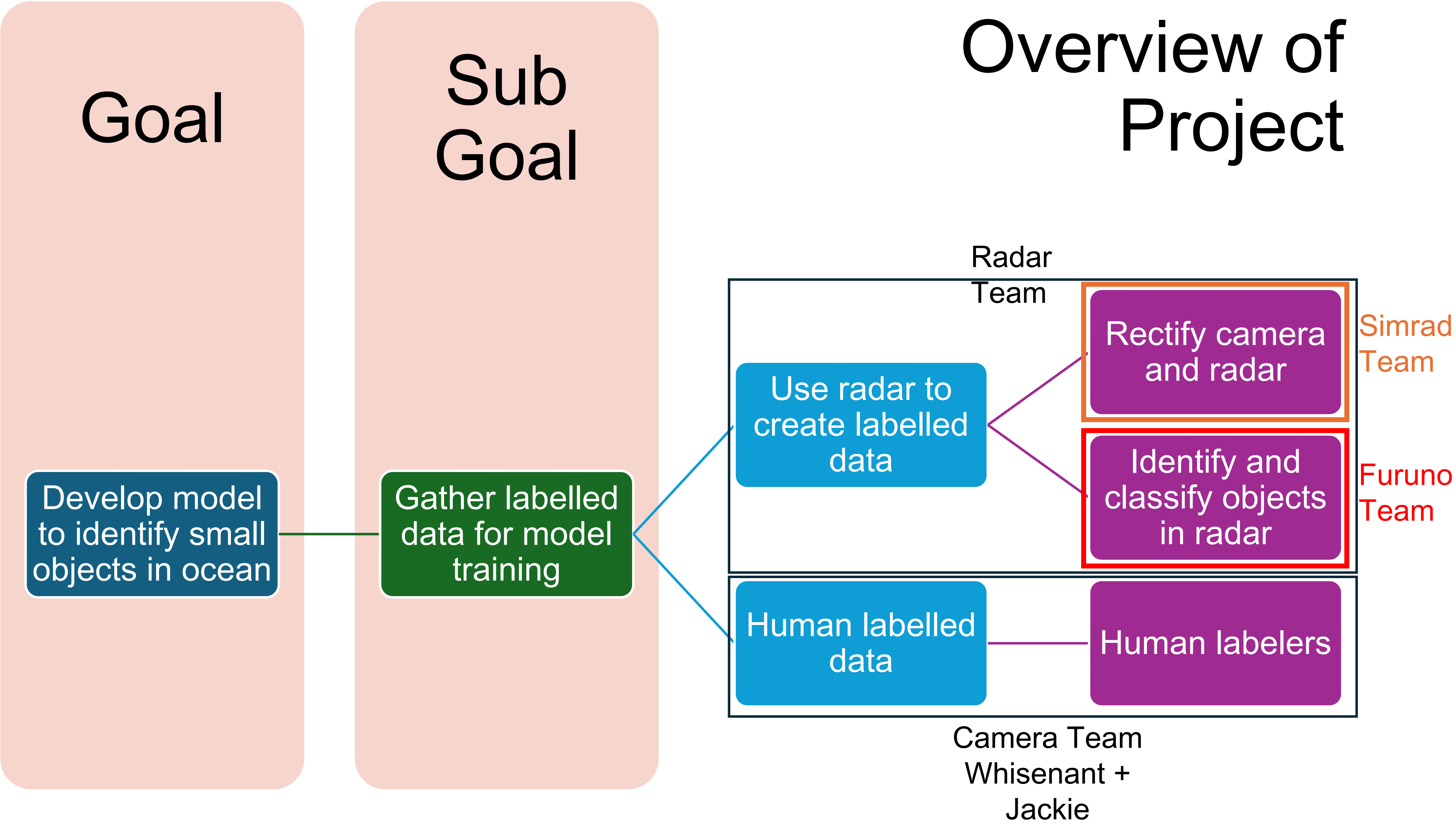


(d) Radar overlay

The Furuno Team

(Furuno Team uses Furuno Radar, 24”)

Overview of Project



Overview of Furuno Team

Goal

- Identify and Classify Objects in Radar sweeps

Sub Goals

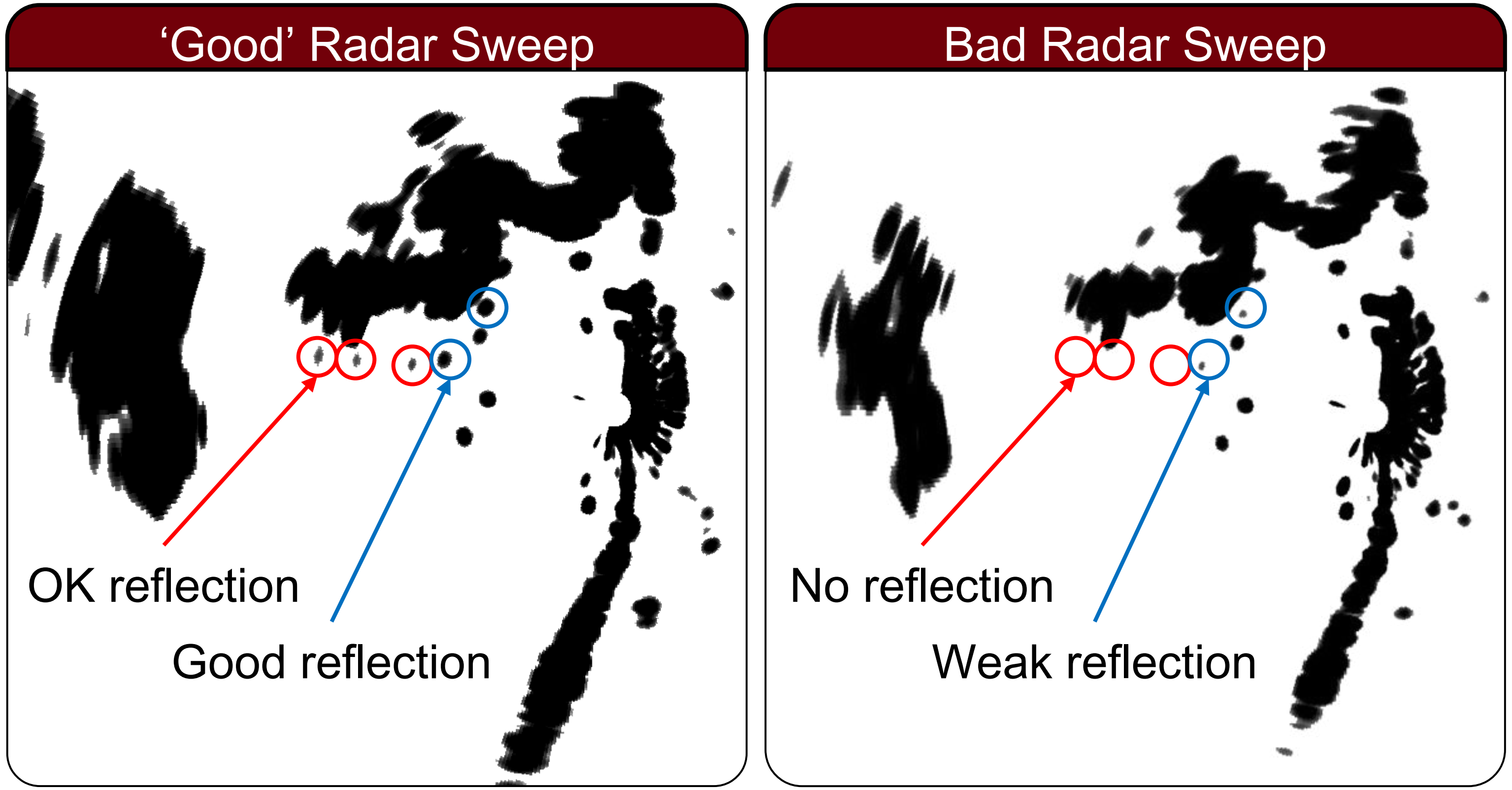
- Gather data for training/testing
- Develop models for tracking

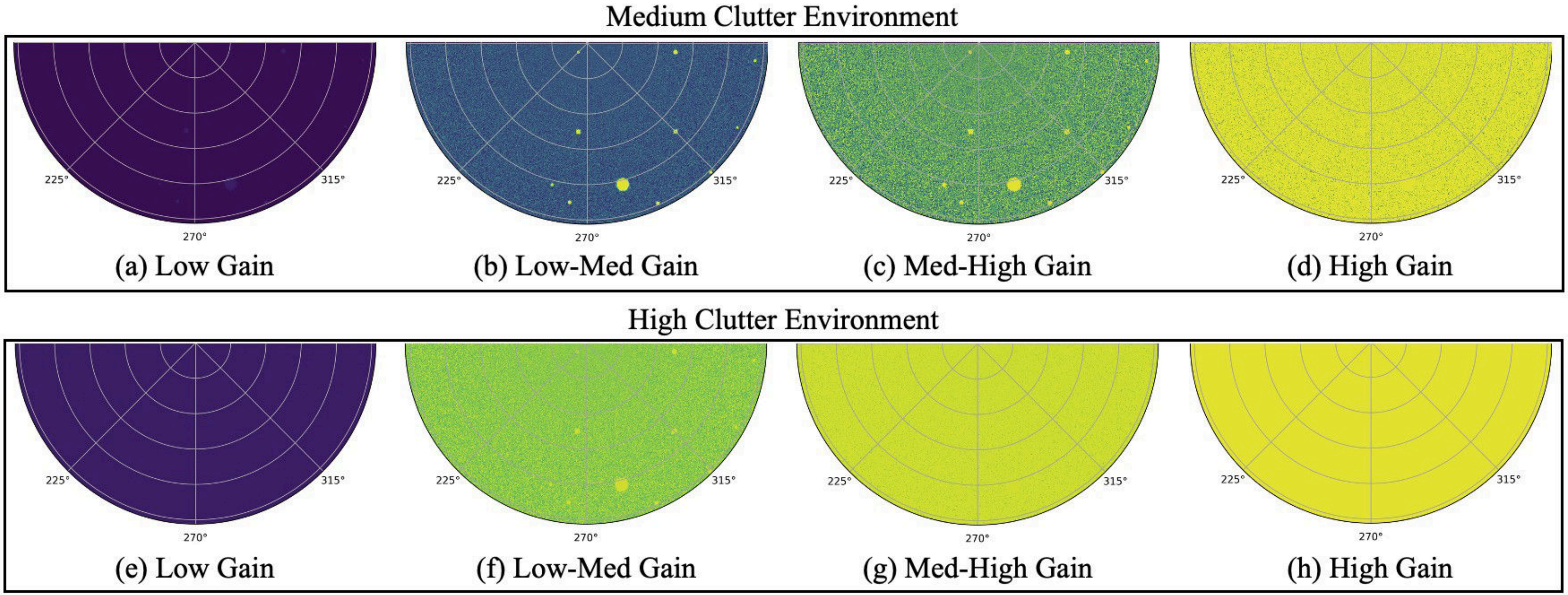
Specific Goals

- Collect at multiple sites
- Test approaches (multiple type)

The Radar Problem

- Oftentimes the sweeps will not capture every object due to a variety of issues.

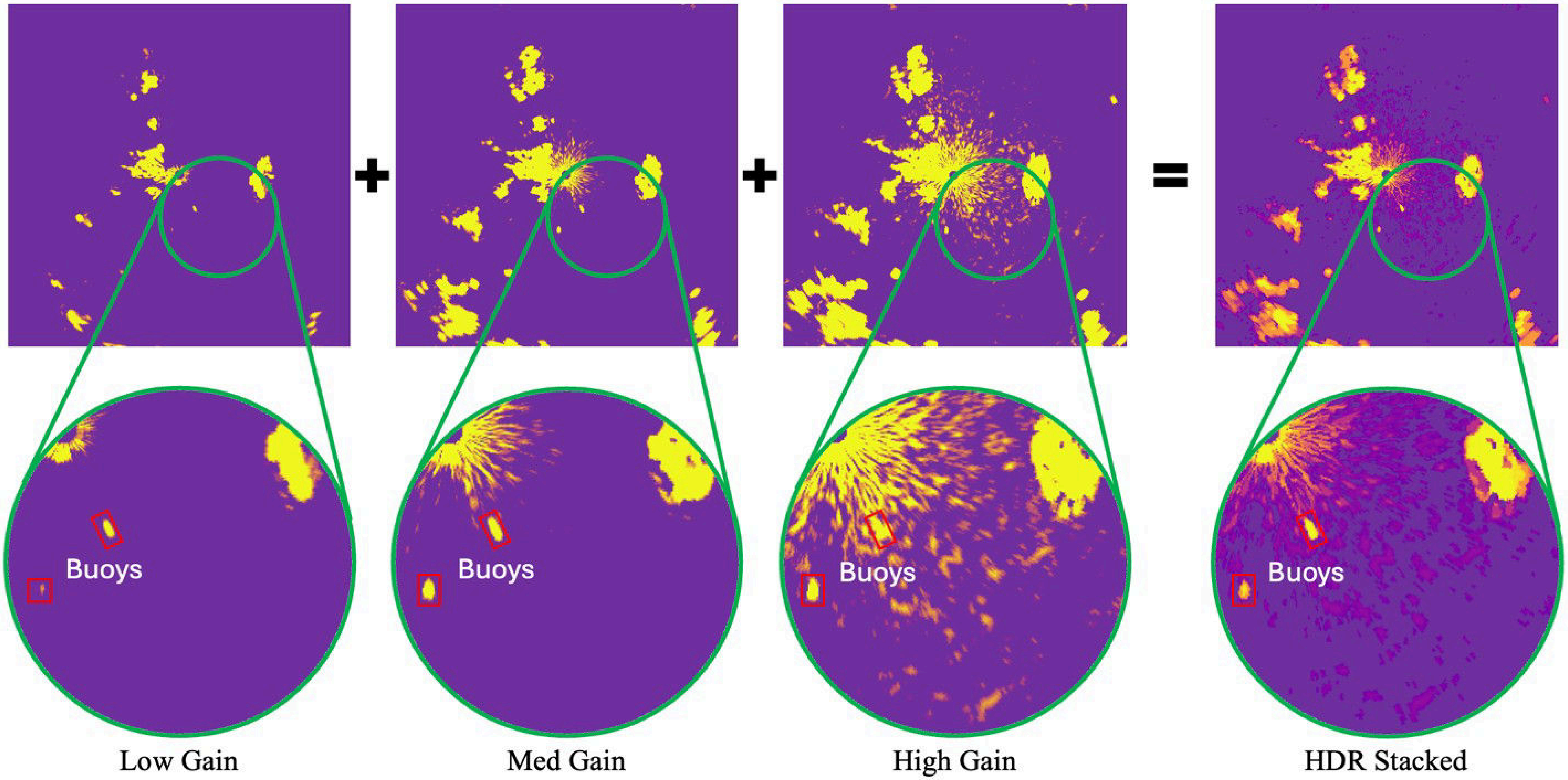




HDR – High Dynamic Range

Sometimes objects are not detectable at all Gains. So, we combine frames to improve P_d

HDR Result



Buoys
Buoys
Buoys
Buoys
Low Gain
Med Gain
High Gain
HDR Stacked

Current Pipeline

Collect Data → Land v Object → Moving v Stationary → Tracking

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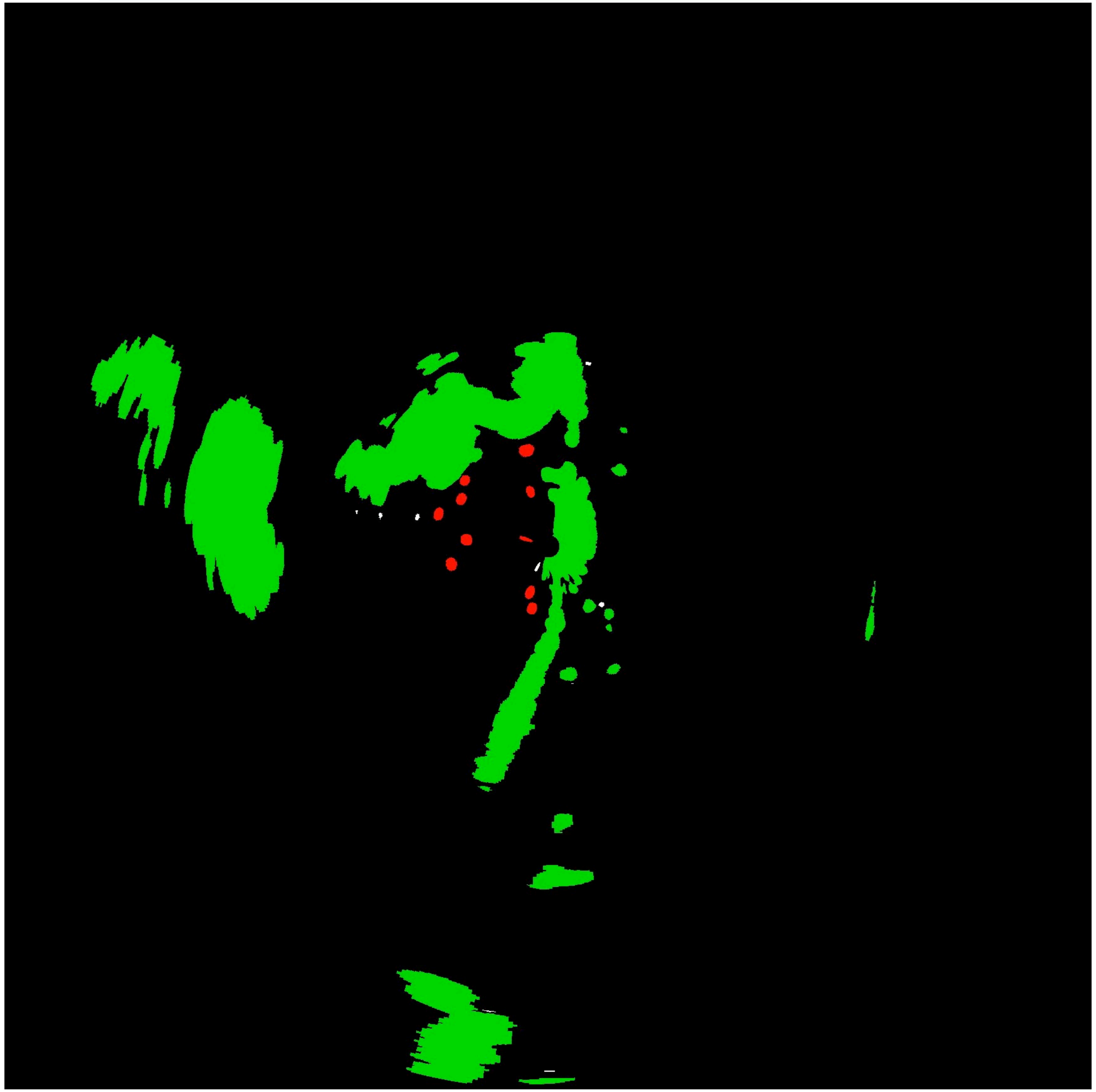
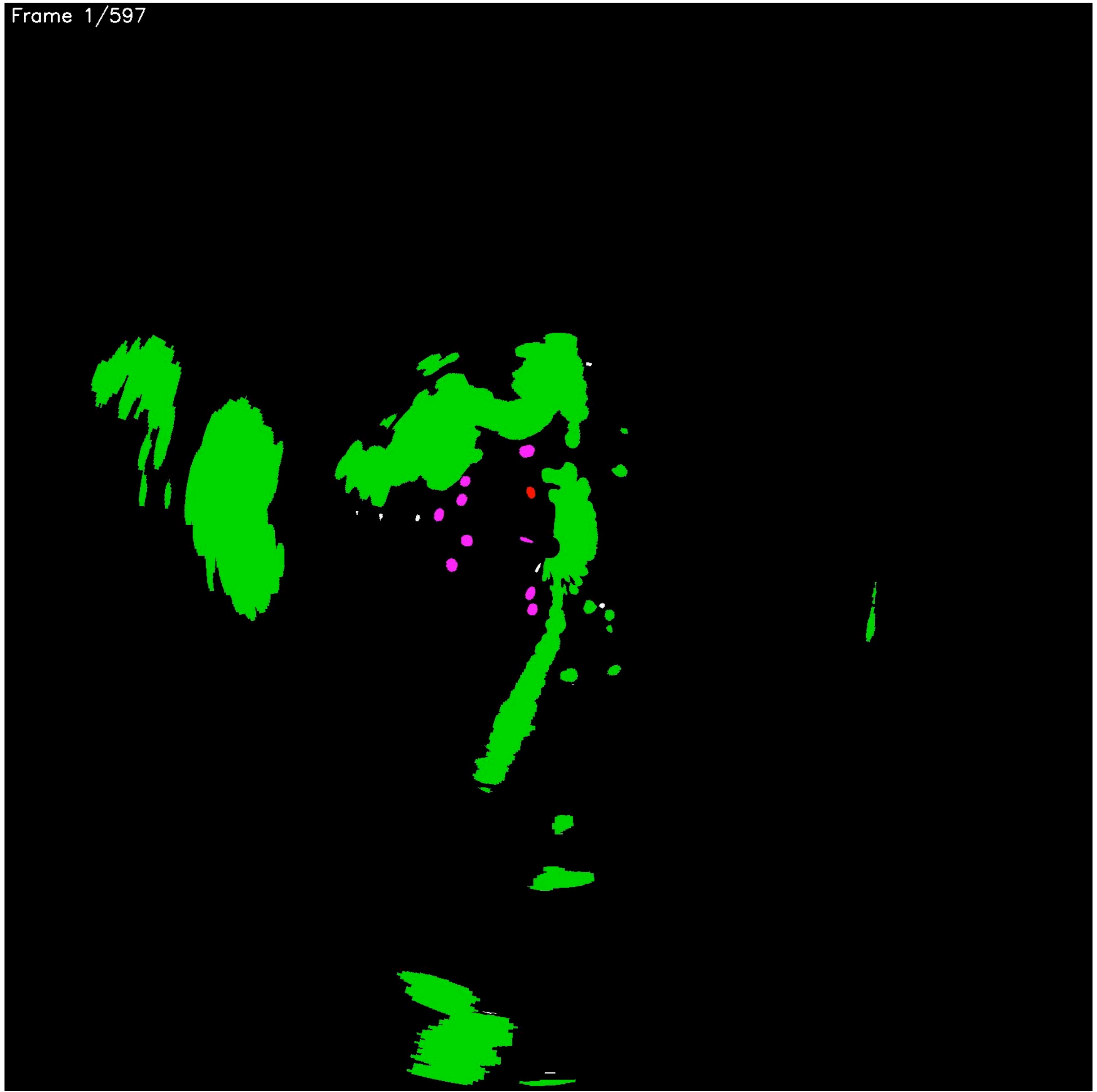


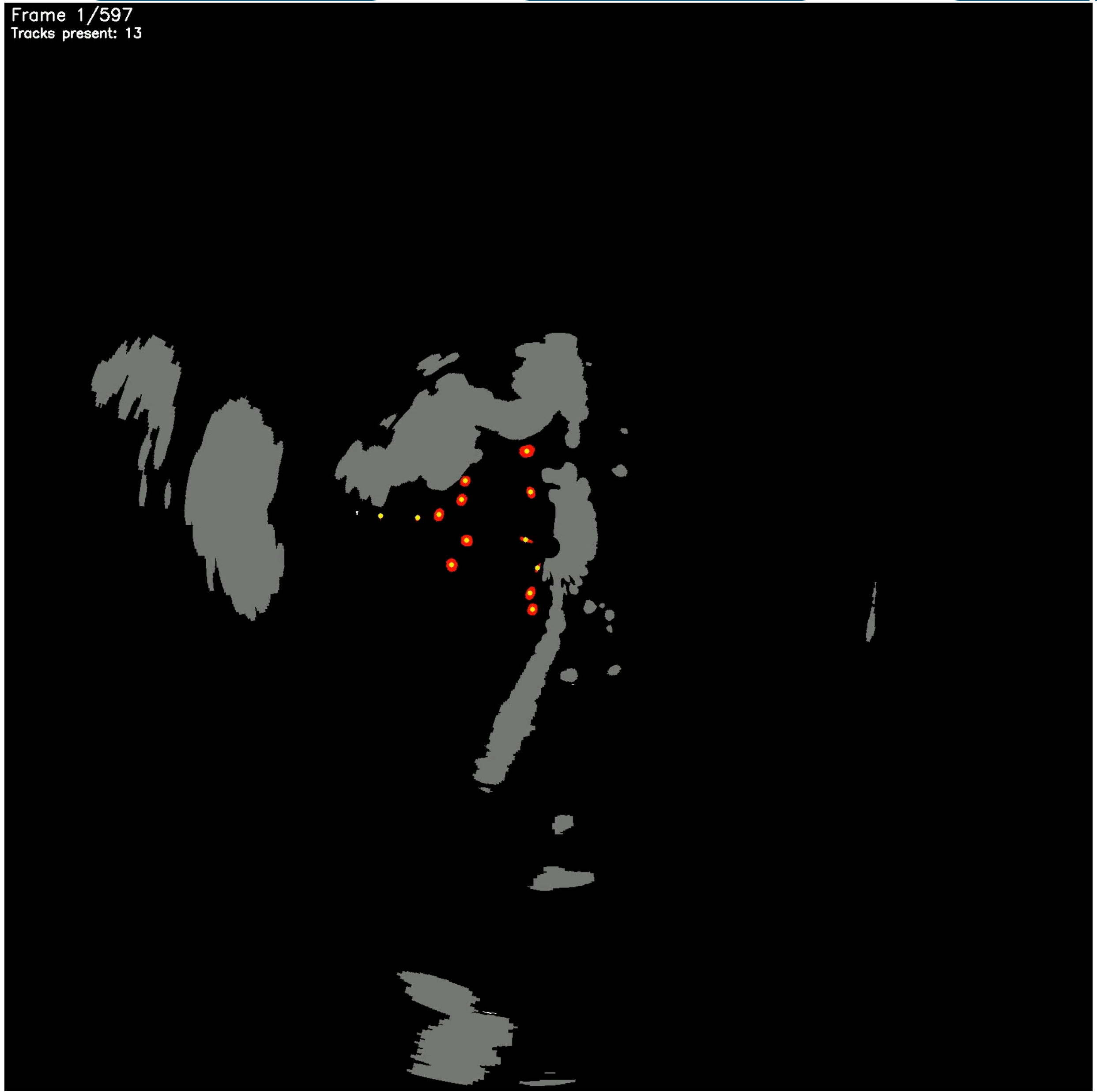
Current Pipeline

Collect Data → Land v Object → Moving v Stationar y → Tracking



Current Pipeline

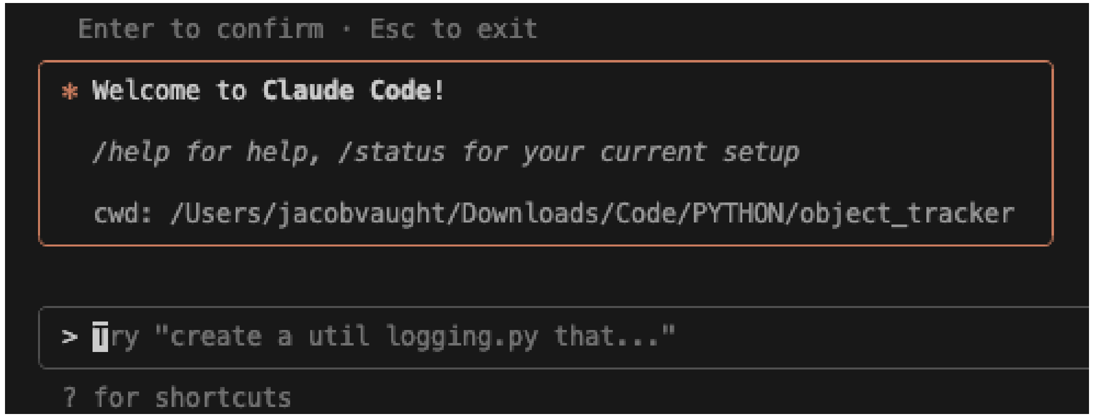
Collect Data → Land v Object → Moving v Stationary → Tracking

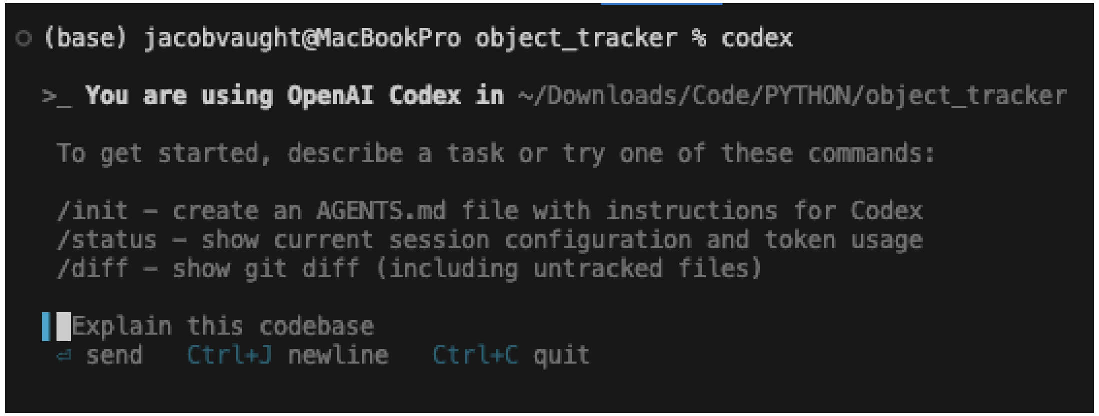


Agentic Code Summary

Paid Options

- Anthropic’s Claude Code
  - Best at getting code done; generally organized in its coding
    - npm install -g @anthropic-ai/claude-code
- OpenAI Codex CLI
  - Alright. Better at doing tasks than Gemini and Qwen, but worse than Claude
    - npm install -g @openai/codex





Agentic Code Summary

Free Options

- Google Gemini CLI
  - 1,000 requests free per day
  - Good at open-ended problems
    - npm install -g @google/gemini-cli
- Alibaba Qwen Code
  - 2,000 requests free per day
  - Better organization than Gemini, but worse at open-ended.
    - npm install -g @qwen-code/qwen-code

Work Plan

Cancilla

- Algorithm Exploration
  - Inventing new ways to find targets in our radar data without using any labels.
- Presentation ~1wk
  - Some implemented algorithms and data collection

Kirk

- Data & Infrastructure Development
  - Improving the quality of our data and building needed tools.
- Presentation ~2wk
  - Just data collection, preferably some augmentation work

Cancilla: Unsupervised Baseline

- Can we detect and track targets with traditional, non-ML techniques?
- This should give us a simple, fast baseline to compare all our complex deep learning models against.

- Write a script that applies the DBSCAN clustering algorithm to find dense groups of radar hits in a noisy scan.
- Investigate using Optical Flow to track these clusters between frames.

Cancilla: Point Clouds

- Treat our radar data like a self-driving car sees the world; as a 3D "point cloud."
- This approach might be more natural for sparse radar data than 2D images. It's a ‘risky’ path that could lead to a breakthrough.

- Write a script to reformat our radar data into a point cloud format.
- Investigate how to train a simple, self-supervised PointNet model to learn the structure of our radar scans.

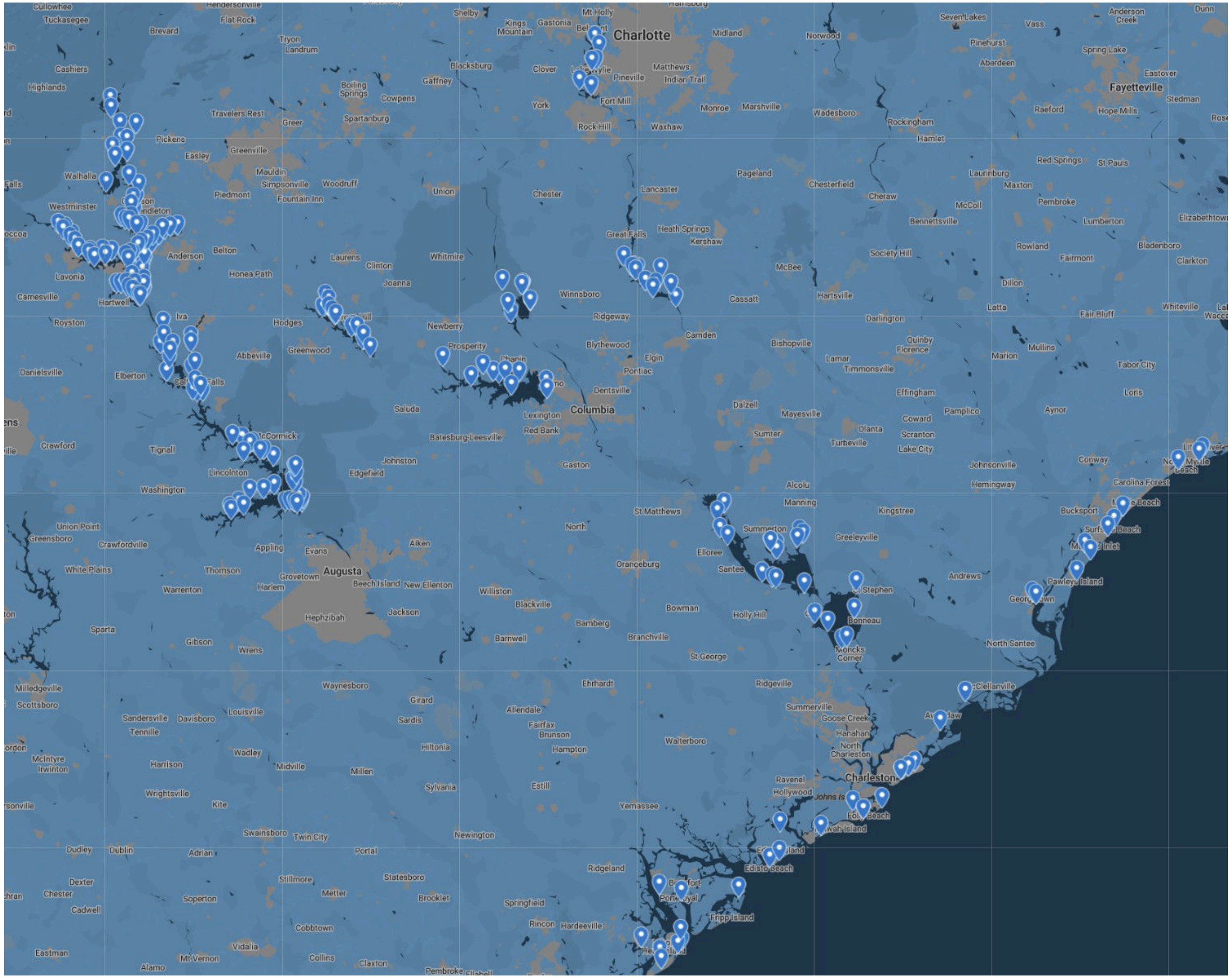
Cancilla: Anomaly Detector

- Frame the problem differently: a boat is simply an "anomaly" in a normal background of sea clutter.
- This approach could turn the problem into a classic, solvable one. It could be a very fast and effective way to flag potential targets.

- Write a script to train an Isolation Forest or One-Class SVM model on radar scans containing only sea clutter (this will need Charleston data, or big lakes and high gain)
- Apply the trained model to new scans containing boats and see if they are correctly flagged as anomalies.

Kirk: Collect Data

- Collect data at different locations along the lake and test scripts as needed.
- Collect metadata(i.e. GPS, heading, weather, wind)



Kirk: Clean Data

- Make our raw radar signal as clean as possible before it ever gets to a fancy model.
- A cleaner signal ***CAN*** dramatically improves the performance of every single model we build. It can also make the model worse if done poorly.

- Write a script to implement and compare at least two different CFAR (Constant False Alarm Rate) algorithms on our noisiest data. (try on high gain data first)
- Your goal is to find the absolute best clutter removal technique for our specific radar and environment

Kirk: Data Augmentation

- We can't be on the lake 24/7, but we can use software to create more data. This should allow us to artificially increase the size and variety of our dataset.
- More diverse training data helps our self-supervised models (like the MAE) learn faster and become more robust.

- Create a "Radar Augmentation" library.
- Write a script that can take one radar scan and generate 10 new, "fake" versions by applying rotations, simulated noise, and other transformations.

Kirk: Sensor Fusion & Non-Littoral data

- Our biggest long-term goal is fusing radar and camera data, but we can't do it without paired, synchronized data. This path is about solving that problem permanently.
- Additionally, all of our data is shore bound. This is a major downside to many of the algorithms utilized.

- Design and build a system to capture synchronized data. This involves physically mounting a camera on the radar rig. (I’ve used an iPhone before)
- If the pontoon is ready to work on, work on integrating a quick connect system for mounting the radar to it. Brad is working on the pontoon.
- You can collaborate closely with the Simrad Team, who are working on camera calibration and rectification.

By end of semester – Final goals

- Have 2-3 abstracts ready to submit to conferences regarding the work done
- Have open dataset started – at least v0.1 and final version of how to collect data.
- Have new version of HDR(?) implemented (uses pulse length)
- Have better(maybe 20% better?(relative)) tracking/classification model than today
- More baseline comparison algorithms
- Augmented Data Pipeline completed

Questions?